

Sayali A. Alatk

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EDUCATION

University of Wisconsin-Madison, Madison, Wisconsin USA

- Ph.D. (Computer Science), August, 2025
- Ph.D. Thesis: Biologically interpretable latent representations in single-cell multimodalities
- Committee: Daifeng Wang (advisor), Fred Sala, Yudong Chen, André Sousa, Panos Roussos

Stony Brook University, Stony Brook, New York USA

- M.S. (Computer Science), May, 2020
- Master's Thesis: Detecting Smart Home Activity through Network Traffic Signatures.
- Committee: Samir Das (advisor), Amir Rahmati, Michalis Polychronakis, Vasudevan Nand

University of Pune, Pune, India

B.E. (Computer Engineering), May, 2018

PROFESSIONAL EXPERIENCE

Stanford University, Stanford, CA, USA

Postdoctoral Scholar, Pritchard Lab

September, 2025 - present

- Developing computational methods for perturb-seq data to study gene regulatory networks.

UW-Madison, Madison, WI, USA

Research Assistant, Daifeng Wang Lab & Waisman Center

August, 2021 - August, 2025

- Developing interpretable machine learning methods for single-cell genomics (e.g., scRNA-seq, scATAC-seq), spatial transcriptomics and genotype data
- Assisted on several NIH grant proposals

Siemens Corporate Research, Princeton, NJ, USA

Research intern, Cybersecurity Research Group

May, 2019 - August, 2019

- Implemented an OCR-based homograph detection tool from literature for domain service monitoring
- Implemented new features for Siemens threat news portal

PUBLICATIONS

Submitted/under-review

1. Dynamic landscapes of gene regulatory networks in early mammalian neurogenesis: Insights into brain evolution and disorder risk, *submitted*, 2025, link
Kalpana Hanthanan Arachchilage, Ryan D Risgaard, Jie Sheng, Pubudu Kumarage, Jerome J Choi, **Sayali Anil Alatk**, Chirag Gupta, Xinyu Zhao, André MM Sousa, Daifeng Wang,
2. NeuroTD: A Time-Frequency Based Multimodal Learning Approach to Analyze Time Delays in Neural Activities, *submitted*, 2025, link code
Xiang Huang, Noah Cohen Kalafut, **Sayali Alatk**, Athan Z. Li, Qiping Dong, Qiang Chang, Daifeng Wang,

Peer-reviewed/conference papers

1. Personalized Single-cell Transcriptomics Reveals Molecular Diversity in Alzheimer's Disease, *Nature Communications*, 2026, link code
Pramod Bharadwaj Chandrashekar*, **Sayali Anil Alatk***, Noah Cohen Kalafut*, Ting

Jin*, Chirag Gupta, Ryan Burzak, Xiang Huang, Shuang Liu, Athan Z. Li, PsychAD Consortium, Kiran Girdhar, Georgios Voloudakis, Gabriel E. Hoffman, Jaroslav Bendl, John F. Fullard, Donghoon Lee, Panos Roussos#, Daifeng Wang#

2. ARTEMIS integrates autoencoders and Schrödinger Bridges to predict continuous dynamics of gene expression, cell population and perturbation from time-series single-cell data, *Bioinformatics*, ISMB/ECCB 2025 (acceptance rate 12%), link code
Sayali Anil Alatkarak, Daifeng Wang,
3. CMOT: Cross-Modality Optimal Transport for multimodal inference, *Genome Biology*, 24, 163, 2023,link code
Sayali Anil Alatkarak, Daifeng Wang,
4. DeepGAMI: Deep biologically guided auxiliary learning for multimodal integration and imputation to improve phenotype prediction, *Genome Medicine* 15, 88, 2023,link code
Pramod Bharadwaj Chandrashekar, **Sayali Alatkarak**, Jiebiao Wang, Gabriel E. Hoffman, Chenfeng He, Ting Jin, Saniya Khullar, Jaroslav Bendl, John F. Fullard, Panagiotis Roussos, Daifeng Wang,
5. Single-cell network biology characterizes cell-type gene regulation for drug repurposing and phenotype prediction in Alzheimer’s disease, *PLoS Computational Biology*, 18(7): e1010287, 2022
Chirag Gupta, Jieli Xu, Ting Jin, Saniya Khullar, Xiaoyu Liu, **Sayali Alatkarak**, Feixiong Cheng, Daifeng Wang, link

TEACHING EXPERIENCE

UW-Madison, Madison, WI, USA
Teaching Assistant-Intro to Python

August, 2020 - May, 2021

POSTERS/TALKS

Posters

- Research in Computational Molecular Biology (RECOMB) ’21
- International Conference on Intelligent Systems for Molecular Biology (ISMB) ’22

Talks

- ISMB/European Conference on Computational Biology (ECCB) ’25
- RECOMB/ISCB Conference on Regulatory & Systems Genomics with DREAM Challenges (RSG-DREAM) ’23

MENTORING

- Abhinav Nandwani, *B.S. in Computer Engineering, UW-Madison* (Spring ’25 - Summer ’25)
- Ryan Burczak, *M.S. in Biomedical Data Science, UW-Madison* (Spring ’24 - Fall ’24)

HONORS AND AWARDS

UW-Madison CS Summer Research Fellowship
Usenix Security’21 Diversity Grant
Accepted into NSF Sponsored GREPSEC (Workshop for Underrepresented Groups in Security and Privacy) V Workshop’21

SKILLS

- Languages: Python, R
- Packages (ordered by proficiency): PyTorch, PyTorch Geometric, JAX, DGL
- Applications: Visual Studio Code, Anaconda, RStudio, Cytoscape
- Operating Systems: Ubuntu, Windows